

Problem 1

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Problem 1 (a) Let $f(x) = \frac{x-1}{x^2-5x}$. Then $f(2) = -\frac{1}{6}$ and $f(6) = \frac{5}{6}$, but there is no value of c between 2 and 6 for which $f(c) = 0$. Does this fact violate the Intermediate Value Theorem?

Explanation. It does not violate the Intermediate Value Theorem. f is not continuous at 5 so the conditions of the IVT do not hold and therefore the IVT does not apply.

(b) *True or False: At some time since you were born your weight in pounds exactly equaled your height in inches.*

Explanation. True: if $w(t)$ represents your weight in pounds at time t and $h(t)$ represents your height in inches at time t , then w and h are both continuous functions. This implies $w - h$ is also continuous. If $t = 0$ is the moment you were born and $t = T_0$ is the present time, then $w(0) - h(0) < 0$ and $w(T_0) - h(T_0) > 0$. Hence by the Intermediate Value Theorem there is a point in the past, t , when $w(t) - h(t) = 0$ and therefore your weight in pounds equaled your height in inches.